

BuildSite 360

Construction Site Monitoring Platform

Comprehensive Project Report

Project Status	IMPLEMENTATION COMPLETE
Report Date	September 13, 2026
Version	1.0

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1. Project Implementation Overview

BuildSite 360 is a comprehensive construction site monitoring platform designed to streamline operations, enhance safety compliance, and provide real-time visibility into project progress. The platform integrates AI-powered safety detection, biometric attendance tracking, real-time communication, and detailed site analytics into a cohesive system.

Component	Status	Details
Frontend (React 18)	✓ Complete	Vite-based SPA with Context API
Backend (Express.js)	✓ Complete	40+ REST endpoints with Socket.io
AI Service (FastAPI)	✓ Complete	YOLOv8 + DeepFace integration
Database (PostgreSQL)	✓ Complete	14 Prisma models with full schema
Real-time System	✓ Complete	Socket.io for WebSocket events

2. Technology Stack

Frontend Technologies

Technology	Version	Purpose
React	18.2.0	UI framework
Vite	4.4.0	Build tool
Axios	1.4.0	HTTP client
Socket.io-client	4.5.0	Real-time communication
Recharts	2.10.0	Data visualization

Backend Technologies

Technology	Version	Purpose
Express.js	4.18.2	Web framework
Prisma	5.0.0	ORM
PostgreSQL	15	Database
Socket.io	4.5.0	Real-time events
JWT	9.0.0	Authentication
Bcrypt	5.1.0	Password hashing
AWS SDK v3	3.549.0	S3 storage
node-cron	3.0.2	Job scheduling
PDFKit	0.13.0	PDF generation
Nodemailer	6.9.0	Email service

AI Service Technologies

Technology	Version	Purpose
FastAPI	0.104.1	AI service framework
PyTorch	2.0.1	Deep learning
Ultralytics YOLOv8s	8.0.0	PPE detection
DeepFace	0.0.71	Facial recognition
OpenCV	4.8.1.78	Image processing
NumPy	1.24.3	Numerical computing
scikit-learn	1.3.0	ML utilities

3. Architecture & Services

Monorepo Structure: BuildSite 360 is organized as a single Git repository containing three independent services:

client/ - React 18 frontend built with Vite. Uses Context API for state management, no Redux/Zustand. Communicates with backend via Axios with JWT interceptor and Socket.io for real-time updates.

server/ - Express.js API built with CommonJS. Manages 40+ REST endpoints nested by project ID. Integrates Prisma ORM, Socket.io for real-time events, AWS S3 integration, and FFmpeg for camera streaming.

ai-service/ - FastAPI microservice bound to localhost only. Implements YOLOv8s for PPE detection and DeepFace/ArcFace for facial recognition. Communicates with Node backend via X-Internal-Token authenticated endpoints.

Authentication Flow: JWT access tokens (24h) + httpOnly refresh cookies (7d). Two-layer authorization: role check (ADMIN/ENGINEER/CLIENT) + project membership verification. Passwords hashed with bcrypt (cost factor 12).

Database Schema - 14 Core Models:

Model	Key Fields	Relationships
User	email, role, passwordHash	Projects, Workers, Messages
Project	name, status, budget, GPS	Engineers, Updates, Materials
ProjectEngineer	projectId, engineerId	Assigned to projects
SiteUpdate	description, mediaUrls	Per project/engineer
MaterialEntry	name, quantity, unitCost	Inventory tracking
Camera	name, zone, rtspUrl	Zone monitoring
SafetyAlert	violationType, confidence	NO_HELMET/NO_VEST violations
Worker	name, faceEmbedding	Biometric data
AttendanceRecord	checkInTime, confidence	Face-matched tracking
ChatMessage	content, fileUrl	Project communications
Subscription	plan, status, provider	BASIC/PRO/ENTERPRISE
Report	reportType, pdfUrl	DAILY/WEEKLY generation
MediaAsset	url, source, caption	Shared project library

4. Core Features Implemented

✓ User & Project Management

Multi-role access (Admin, Engineer, Client) with granular project-level permissions. Project creation, engineer assignment, and budget tracking. Real-time project status updates (PLANNING → ACTIVE → ON_HOLD → COMPLETED).

✓ Safety Monitoring & Alerts

AI-powered PPE detection (helmets, vests) via YOLOv8s. Real-time safety violation detection with confidence scoring. Configurable alert thresholds and cooldown periods. Historical safety alert logs with video frame snapshots.

✓ Biometric Attendance Tracking

Worker facial recognition enrollment and face-embedding storage. Automatic check-in on camera detection. Match confidence scoring. Daily attendance records with unique worker-date constraints.

✓ Site Updates & Documentation

Engineer-created daily/weekly progress updates with media attachments. Image and video uploads to S3/MinIO. MediaType tracking (IMAGE, VIDEO, MIXED). Chronological site documentation.

✓ Material & Inventory Management

Material receipt and consumption logging. Per-unit cost tracking with Decimal precision. Category and type classification. Project-based inventory visibility.

✓ Real-time Communication

Project-scoped chat with Socket.io WebSocket events. Media file sharing with optional MediaAsset library linking. User presence awareness. Message persistence to database.

✓ Analytics & Reporting

Daily and weekly report generation via node-cron. PDF report creation with PDFKit. Safety metrics and attendance summaries. Material cost analytics. Scheduled email delivery via Nodemailer.

✓ Multi-Camera System

Zone-based camera management (ENTRANCE, WORK_AREA, STORAGE). RTSP stream support with FFmpeg relay. Camera status tracking (ONLINE/OFFLINE). Real-time safety alerts linked to specific cameras.

✓ Media & Asset Management

Centralized project media library. Auto-recording from camera clips (CLIP_RECORDING source). Manual uploads (MANUAL_UPLOAD source). Media sharing in chat with library linkage.

✓ Payment & Subscriptions

Multi-provider support (EasyPaisa, JazzCash, Stripe). Tiered subscription plans (BASIC, PRO, ENTERPRISE). Webhook signature verification for payment providers. Subscription lifecycle management.

5. Deployment & DevOps

Environment Configuration: Each service maintains its own .env file (not committed): server/.env, client/.env.local, ai-service/.env. Configuration includes database URLs, JWT secrets, API keys, and service endpoints.

Testing Infrastructure: Postman/Newman E2E test suite (40+ API endpoints), Python integration tests for AI service, manual QA test scripts. Database seeding via comprehensive seed scripts.

Security Hardening: Two-layer authorization (role + project membership). Fail-closed authentication design. X-Internal-Token for service boundary protection. Camera source validation before system calls. Webhook signature verification over raw request bodies.

Scaling Considerations: PostgreSQL connection pooling ready. S3 integration for unlimited file storage. Socket.io scalable to multiple server instances. FFmpeg process management for concurrent streams.

6. Recommendations & Future Work

Short-term Enhancements:

- API rate limiting on authentication endpoints
- Enhanced error handling and user feedback
- Performance monitoring and logging integration
- Mobile-responsive UI refinements

Medium-term Roadmap:

- Advanced analytics dashboard with predictive insights
- Machine learning model improvements (crowd detection, fatigue detection)
- Multi-site dashboard and cross-project analytics
- Mobile native apps (iOS/Android)
- Integration with construction management software

Long-term Vision:

- AI-powered resource optimization
- Drone/aerial imagery integration
- Augmented reality site visualization
- Blockchain-based compliance tracking
- Integration with IoT sensors (temperature, humidity, air quality)

Operational Excellence:

- Implement comprehensive logging and monitoring (ELK, DataDog, etc.)
- Set up CI/CD pipeline (GitHub Actions, GitLab CI)
- Container orchestration (Docker, Kubernetes)
- Disaster recovery and backup procedures
- Performance optimization for large-scale deployments